

## **EXPERIMENT-24 Diet Suggestion Based on Disease**

### **Aim**

To develop a Prolog-based dieting system that suggests suitable food based on disease.

### **Objectives**

- To represent diseases and recommended foods.
- To use rules for diet recommendations.
- To demonstrate knowledge-based reasoning.

### **Algorithm**

1. Define diseases.
2. Define foods suitable for each disease.
3. Create a rule to recommend food.
4. Query the system with a disease.
5. Display recommended foods.

### **Flowchart**

START

|

v

Enter disease

|

v

Check disease facts

|

v

Find recommended food

|

v

Display diet

|

v

STOP

## **Program**

% Diet Suggestion System

diet(diabetes, vegetables).

diet(diabetes, whole\_grains).

diet(diabetes, nuts).

diet(hypertension, fruits).

diet(hypertension, vegetables).

diet(hypertension, low\_salt\_food).

diet(anemia, spinach).

diet(anemia, beans).

diet(anemia, iron\_rich\_food).

diet(obesity, vegetables).

diet(obesity, fruits).

diet(obesity, low\_calorie\_food).

recommend(Disease, Food) :-

diet(Disease, Food).

## Queries

?- recommend(diabetes, Food).

Output:

Food = vegetables ;

Food = whole\_grains ;

Food = nuts ;

false.

?- recommend(anemia, Food).

Output:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ≡ HelloWorld.pl ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | I/O AI Agent                                                                                                                                                                                                                   |
| <pre>1 % Diet Suggestion System 2 3 % Diet facts 4 diet(diabetes, vegetables). 5 diet(diabetes, whole_grains). 6 diet(diabetes, nuts). 7 8 diet(hypertension, fruits). 9 diet(hypertension, vegetables). 10 diet(hypertension, low_salt_food). 11 12 diet(anemia, spinach). 13 diet(anemia, beans). 14 diet(anemia, iron_rich_food). 15 16 diet(obesity, vegetables). 17 diet(obesity, fruits). 18 diet(obesity, low_calorie_food). 19 20 % Recommendation rule 21 recommend(Disease, Food) :- 22     diet(Disease, Food). 23 24 % Main program 25 main :- 26     write('Diet Recommendations'), nl, 27     write('-----'), nl, 28 29     write('Diabetes: '), 30     recommend(diabetes, Food1),</pre> | <p>STDIN</p> <p>Input for the program ( Optional )</p> <hr/> <p>Output:</p> <p>Diet Recommendations</p> <p>-----</p> <p>Diabetes: vegetables</p> <p>Hypertension: fruits</p> <p>Anemia: spinach</p> <p>Obesity: vegetables</p> |

## Result

The Prolog program successfully suggests foods based on the specified disease.

## Conclusion

The program demonstrates how a rule-based knowledge system can provide basic diet recommendations. It is only an educational example and not a medical diet plan.